MILITARY SPECIFICATION

PRESSURE-HEIGHT SLIDE RULE ML-315/GM

1. SCOPE

1.1 Scope.-This specification covers one type of slide rule, designated Pressure-Height Slide Rule ML-315/GM, which is used to compute radiosonde data.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

L-V-351	Vinyl Chloride Polymer And Copolymer Rigid Sheets

Military

MIL-P-116	Preservation, Methods of
MIL-P-7936	Parts And Equipment, Aeronautical, Preparation For Delivery

STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

Joint Committee on Printing

JCP 0-40	Government Paper Specification Standards

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Preproduction.-This specification makes provisions for preproduction testing.

3.2 Standard sample.-A standard sample of the ML-315/GM shall be furnished to the contractor for making the necessary photographic plates. It shall also be used to determine the minimum acceptable standards of material, design, workmanship and performance not specified. If any conflict arises between the sample and the specification, the specification shall govern.

3.2.1 Similarity to standard sample.-The equipment produced under contract shall be identical in all respects with the standard sample except for those details which are not in conformance with this specification (3.6).

3.3 Materials, parts, and processes.-Materials, parts, and processes shall be as specified herein.

3.3.1 Plastic.-The vinyl chloride plastic used in the circular scale and cursor of the slide rule shall conform to L-V-351, Class 1, clear, with a high polish finish, 0.050 inch thick.

3.3.2 Paper.-The plastic coated paper shall be white and shall meet the requirements of JCP 0-40.

3.3.3 Base.-The base shall be fabricated of fiberboard material Masonite, or equal.

3.4 Performance requirements.-The components of the slide rule shall retain the specified degree of flatness (3.8.2) and permanency of bonding (3.5.2.1) when subjected to the following conditions:

3.4.1 Service conditions, operating.-The slide rule will be used in shelters with temperatures varying from 30 degrees F to 130 degrees F and relative humidities varying from 0 to 100 percent.

3.4.2 Service conditions, storage.-The slide rule will be stored under temperature conditions ranging from minus 65 degrees F to plus 160 degrees F and relative humidities from 0 to 100 percent.

3.5 Details of components

3.5.1 General.-The slide rule is designed to facilitate the computation of radiosonde data. It shall consist of a rigid base with all scales, formulas, and instructions required for operation printed on

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one side. A movable circular scale and a cursor shall be mounted on the base by means of a swivel. The scales for pressure, temperature, height, and vertical temperature correction shall be identical to those on the standard sample.

3.5.2 The body of each slide rule shall consist of a bottom paper sheet, a fiberboard base, and a top paper sheet imprinted with the height scale.

3.5.2.1 Bonding.-The top and bottom sheets shall be bonded to the fiberboard base. The bonding shall be complete and permanent without noticeable air inclusions. There shall be no distortion of the markings as a result of the bonding or printing.

3.5.3 Swivel.-The design of the swivel, which attaches the cursor and the circular plastic scale to the main body, shall be such that the following requirements are met.

3.5.3.1 Relative motion.-The operator shall be able to move the cursor and the circular scale with respect to the body without undue effort. However, the swivel shall not be so loose that the cursor or circular scale will not stay set in any given position. The swivel shall permit only angular motion between the body and the cursor and circular scale. No translational motion can be tolerated.

3.5.3.1.1 Change of swivel characteristics.-After considerable use, the swivel shall not loosen or tighten enough so that it fails to meet requirement 3.5.3.1.

3.5.3.2 Other swivel considerations.-It shall not be possible to disassemble the swivel assembly and shall not require adjustments after leaving the manufacturer. It shall not require lubrication. Any lubrication required shall be accomplished by the manufacturer at the time of assembly.

3.5.4 Printing.-All material printed shall be dense opaque black. Each line, figure and letter shall be sharp and clean-cut, and of uniform width.

3.6 Identification of product.-Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following nomenclature shall be printed on the slide rule: Pressure-Height Slide Rule ML-315/GM. Figure 1 shows the approximate positioning of the nomenclature.

3.7 Government-loaned property.-When the contract or purchase order so provides, the Government will loan to the contractor, upon his request, one Pressure-Height Slide Rule ML-316/GM.

3.8 Workmanship.-The slide rule, including all parts, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to assure quality of the bonding, and to maintain the high polish finish of the plastic.

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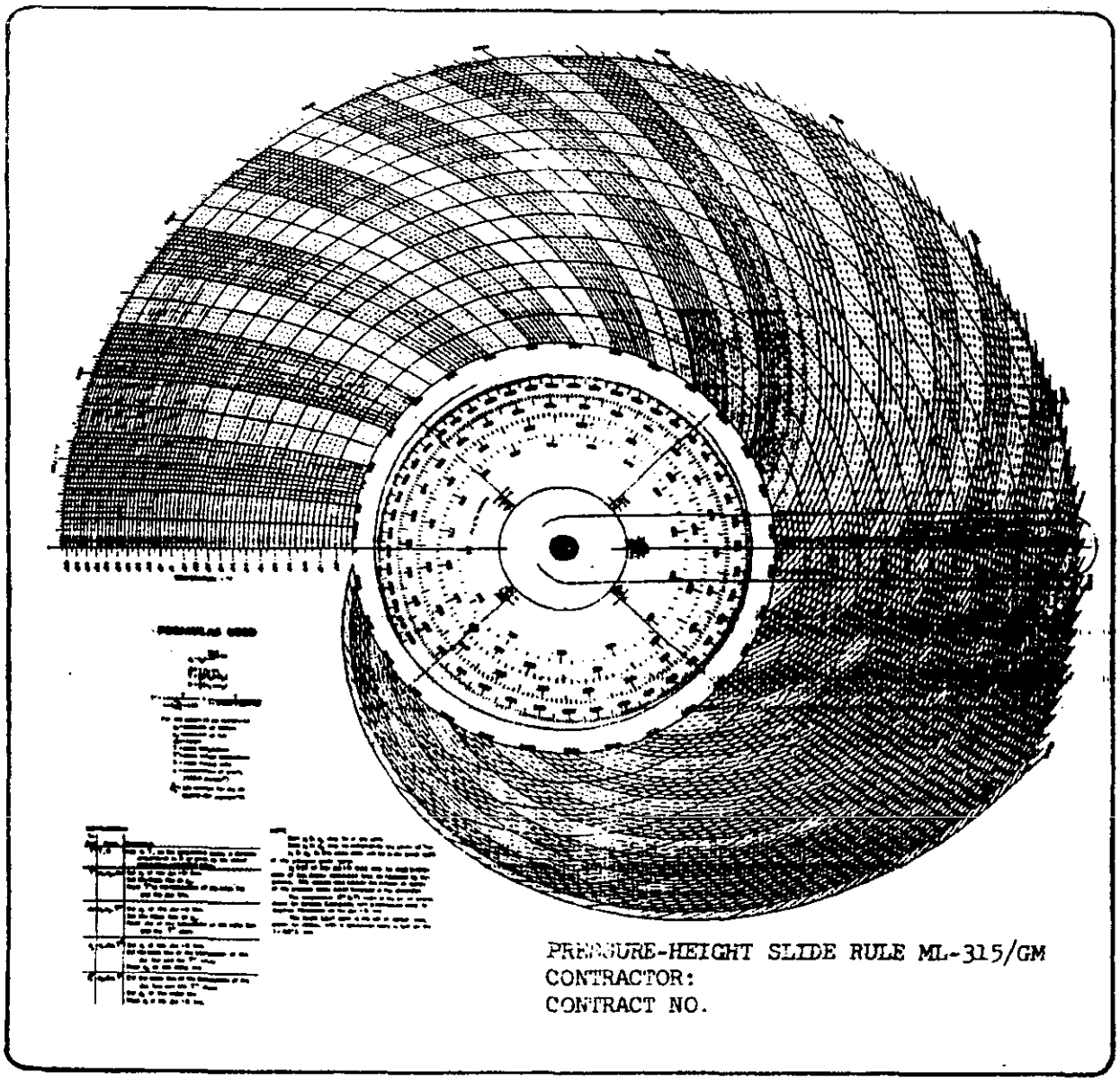


FIGURE 1: PRESSURE-HEIGHT SLIDE RULE ML-315/GM

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3.8.1 Finish.-The finished slide rule shall be free from burrs and rough or sharp edges of metal, wood, and plastic parts alike. All edges and corners shall be rounded. The surfaces shall be free from cracks, seams, bubbles, or raised spots.

3.8.2 Flatness.-There shall be no evidence of waves or ripples in the surface of the slide rule when viewed at an angle of 30 degrees to the line of sight of the observer. The total deviation from a flat surface shall not exceed 1/16 inch either in width or length.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.-The inspection and testing of the slide rule shall be classified as follows:

a. Acceptance tests (4.2)

b. Preproduction tests (4.5).

4.2 Acceptance tests.-Acceptance tests shall conisit of:

a. Individual test (4.2.1)

b. Periodic sampling plan and tests (4.2.2).

4.2.1 Individual test.-Each slide rule shall be subjected to the following test as described under 4.4, Test methods, of this specification: Examination of product (4.4.1).

4.2.2 Periodic sampling plan and tests.-Two slide rules shall be selected at random from each production run of 50 or fraction thereof produced, and subjected to all the tests described under 4.4, Test methods.

4.2.2.1 Rejection and retest.-When one item selected from a production run fails to meet the specification, no items on hand or later produced shall be accepted until the extent and cause of the failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.2.2 Individual test may continue.-For operational reasons, individual tests may be continued pending investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it has been determined that items meet all requirements of the specification.

4.2.3 Defects in items already accepted.-The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.3 Test conditions

4.3.1 Atmospheric conditions.-Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 25 degrees C, plus or minus 10 degrees C, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowances shall be made for the changes in instrument reading.

4.4 Test methods

4.4.1 Examination of product.-Each slide rule shall be inspected to determine compliance with this specification with respect to material, dimensions, markings, workmanship, conformance to figure, bonding, flatness, and finish.

4.4.2 Performance tests

4.4.2.1 The cursor shall be rotated 360 degrees clockwise and 360 degrees counterclockwise for two hundred 360 degree turns. At the end of this test, the torque necessary to turn the cursor shall not be changed by more than 10 percent of its value at the beginning of the test. Also, there shall still be no translational motion between the cursor and the body.

4.4.2.2 The cursor and the circular scale shall be rotated 360 degrees clockwise and 360 degrees counterclockwise for two hundred 360 degree turns. At the end of this test, the torque necessary to turn the cursor and the circular scale shall not be changed by more than 10 percent of its value at the beginning of the test. Also, there shall still be no translational motion between the cursor, the circular scale, and the body.

4.4.3 Service condition tests

4.4.3.1 Operating, high temperature.-The slide rule shall be placed in a temperature chamber, and the internal temperature shall be raised to 130°F. The temperature shall be maintained at plus 130°F for a period of 24 hours. While still at this temperature, the slide rule shall be operated and the results compared with data obtained in 4.4.2.1 and 4.4.2.2. The temperature shall then be reduced to ambient room conditions and the slide rule examined in accordance with 4.4.1.

4.4.3.2 Operating, low temperature.-The slide rule shall be placed in a temperature chamber, and the internal temperature shall be lowered to plus 30°F. The temperature shall be maintained at plus 30°F for a period of 24 hours. While still at this temperature, the slide rule shall be operated and the results compared with data obtained in 4.4.2.1 and 4.4.2.2. The temperature shall then be raised to ambient room conditions and the slide rule examined in accordance with 4.4.1.

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4.4.3.3 Operating, humidity.-The slide rule shall be
a test chamber and the temperature raised to 130°F wi
humidity over a period of two hours. Sufficient mois
the humidity to at least 95% shall then be admitted i
This condition shall be maintained for a period of at
At the end of this period, the slide rule shall be re
examined in accordance with 4.4.1.

pecification

ify the following:

4.4.3.4 Storage, temperature.-The slide rule shall b
a temperature chamber and the temperature reduced to
The temperature shall be maintained at minus 65°F for
one hour, at which time the temperature shall be incr
160°F. for one hour, then returned to ambient conditi
rule shall be removed, and examined per 4.4.1.

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test
loaned property
g desired.

4.5 Preproduction tests

ing Activity:

4.5.1 Preproduction test samples.-The preproduction
shall consist of two models representative of the pro
ment. They shall be tested at a laboratory designate
curing activity or, when so stated in the contract, a
plant under the supervision of the procuring activity

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4.5.2 Preproduction tests.-Preproduction tests shall
tests described under 4.4, Test methods.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.-Preparatio
shall be in accordance with MIL-P-7936. Preservation
shall be in accordance with Method 1C of MIL-P-116.

5.2 Marking for shipment.-Interior packages and exte
containers shall be marked in accordance with MIL-STD
shipment marking nomenclature shall be as follows: P
Slide Rule ML-315/GM.

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6. NOTES

6.1 Intended use.-The slide rule described in this s
is intended for use in computing radiosonde data.

6.2 Ordering data.-Procurement documents should spec

a. Title, number, and date of this specificatio
b. Number of preproduction samples and site of
c. Whether the item in 3.7 shall be Government-
d. Level of preservation, packaging, and packin

e placed in
minus 65°F.
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